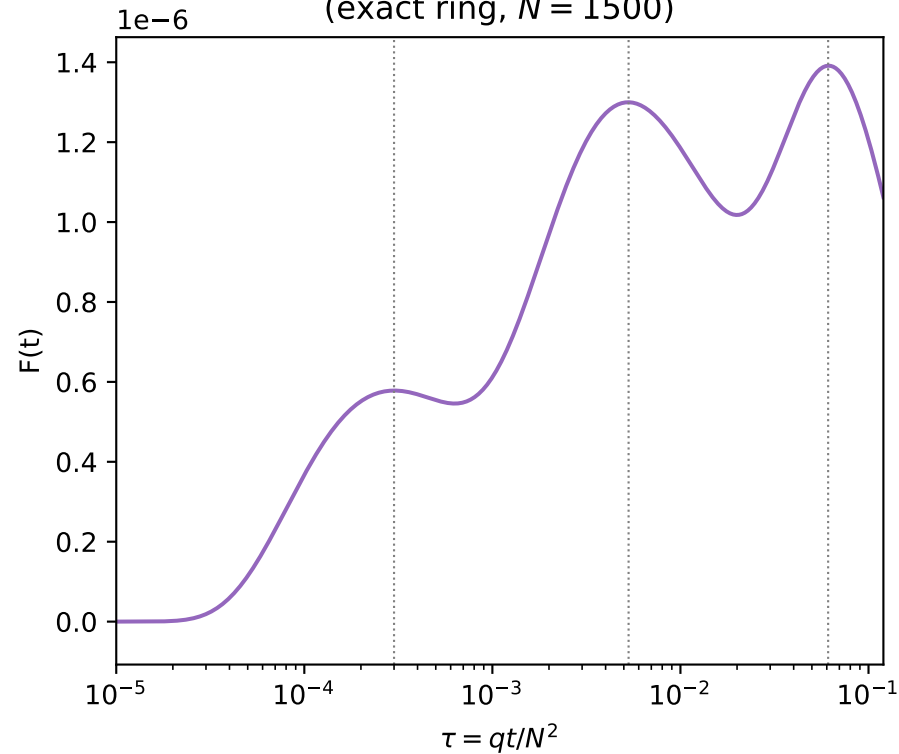
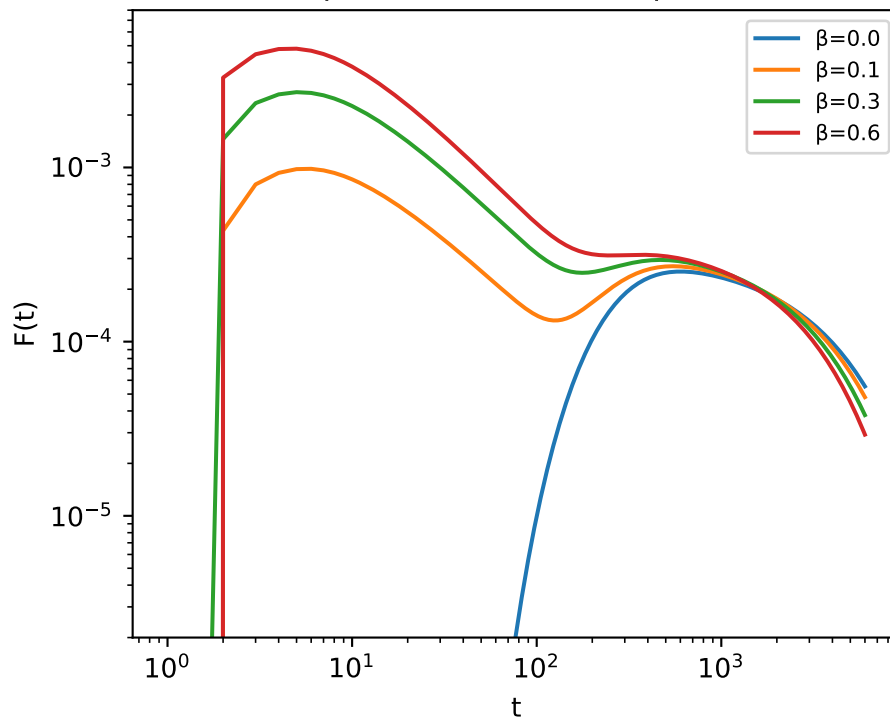


(a) two shortcuts, three peaks
(exact ring, $N = 1500$)



(b) 2D torus (31×31):
capture + late interior peaks



(c) 2D fold: late-peak prominence
 $\rightarrow 0$ at $\beta_c^{2D} \approx 0.68$

